To access to 2d elements : usually by using pair of loops (nested loop)

1	5	2	3	6	7	8	9	4
3	8	4	1	2	5	6	7	9
9	7	1	2	3	4	5	6	8

Q1/ write java program to print all elements inside 2d array

```
public class ArrayExam {
        public static void main(String[] args) {
        String animals [][] = {{"Lion" , "tiger"} , {"elephant"}, {"sanke" , "cat"}  };
        for (int i = 0 ; i<animals.length ; i++)
            for (int j = 0 ; j < animals[i].length ; j++)
                System.out.print(animals[i][j] + " ");

        }
```

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Q2/ Consider the following 2D array named temperatures:
Perform the following:
1. Print out the value of highlighted rows.
2. Print out the value of highlighted columns.
3. Find out the maximum, minimum

12.5	12.2	14.0	13.0	13.2	13.0	12.2
15.2	11.9	11.8	13.2	12.7	14.5	12.4
9.75	11.8	11.9	13.5	12.7	14.7	13.0
14.0	13.2	11.8	14.0	14.3	11.0	11.9

```
public class 2DArrayExam {
public static void main(String[] args) {
double[][] temperatures = {
                {12.5,12.2,14.0,13.0,13.2,13.0,12.2},
                  {15.2,11.9,11.8,13.2,12.7,14.5,12.4},
                  {9.75,11.8,11.9,13.5,12.7,14.7,13.0},
                  {14.0,13.2,11.8,14.0,14.3,11.0,11.9}
                };
        for (int i = 0 ; i < temperatures[0].length ; i++)
            System.out.println(temperatures[0][i]); //1

        for (int i = 0 ; i < temperatures.length ; i++)
            System.out.println(temperatures[i][4]); //2
```

```
double maximum = temperatures[0][0];
double minimum = temperatures[0][0];
for (int i=0 ;i <temperatures.length;i++)
    for (int j = 0 ;j <temperatures[i].length;j++){
        if (temperatures[i][j] > maximum)
            maximum = temperatures[i][j];

        if (temperatures[i][j] < minimum)
            minimum = temperatures[i][j];
    }
  System.out.println(maximum);
  System.out.println(minimum);
    }
```

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Q3/ Write down a Java program to read a 2-dimensional array x[4][4] then perform each of the following:

1. Find the maximum and minimum in the third row
2. Replace the two diagonals with 1

(0,0)	(0,1)	(0,2)	(0,3)
(1,0)	(1,1)	(1,2)	(1,3)
(2,0)	(2,1)	(2,2)	(2,3)
(3,0)	(3,1)	(3,2)	(3,3)

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Q4/ Write a Java program to count the number of occurrences of an element in an 2D array.

```
public class Array2DExample {
        public static void main(String args[]) {
        int counter = 0;
        int numbers[][] = {{1,2,3,4,5,6},{1,2,3,4,5},{9 , 8}};
        for (int i = 0 ; i < numbers.length ; i++)
            for (int j = 0 ; j < numbers[i].length ; j++){
                if (numbers[i][j] == 1) //element want to check
                    counter += 1;
                }
            System.out.println(counter);
            }

        }
```

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Q5/ Write a Java program to print negative numbers in an 2D array.

```
public class Array2DExample {
        public static void main(String args[]) {
        int numbers[][] = {{1,2,3,4,-5,6},{-1,2,-3,4,5},{-9 , -8}};
        for (int i = 0 ; i < numbers.length ; i++)
            for (int j = 0 ; j < numbers[i].length ; j++){
                if (numbers[i][j] < 0) //element want to check
                    System.out.println(numbers[i][j]); }

            }

        }
```

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Q6/ Write a Java program to read of an 2D array then to print the two diagonals elements only.

```
import java.util.Scanner;
public class Array2DExample {
        public static void main(String args[]) {
        Scanner in = new Scanner(System.in);
            int numbers[][] = new int[5][5];
            System.out.println("Write intger values : ");
            for (int i=0 ; i<numbers.length;i++)
               for (int j=0 ; j<numbers[i].length;j++)
                  numbers[i][j] = in.nextInt();

            for (int i=0; i<numbers.length;i++)
               for (int j=0 ; j<numbers[i].length; j++){
                  if ( i == j)
                     System.out.print(numbers[i][j] + " ");}
```

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)

```
            System.out.println("");
            for (int i=0; i<numbers.length;i++)
            for (int j=0 ; j<numbers[i].length; j++){
            if ( i + j == 4)
            System.out.print(numbers[i][j] + " ");}
               }
            }
```

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